212CSE2304	Machine Learning	L	T	P	X	C
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Pre-requisite : NIL
Course Category : Program Core
Course Type : Integrated Course - Theory

COURSE OBJECTIVES:

- To introduce Machine Learning and its benefits
- To understand simple linear regression and its significance.
- To gain knowledge on multiple regression and interpret multiple linear regression coefficients
- To introduce classification and various classification algorithms.
- To provide knowledge on clustering algorithms and various information retrieval models.

COURSE OUTCOMES:

CO1: Understand the origin, uses, concept, types and applications of Machine Learning.

CO2: Understand the mathematics behind learning algorithms and apply simple linear regression on datasets with performance evaluation.

CO3: Design and Evaluate Multiple Regression models on real time datasets.

CO4: Understand and Apply various classification algorithms in Machine Learning and visualize the results.

CO5: Apply various clustering algorithms on real time datasets and analyze the efficiency of each algorithm in different usecase scenarios.

CO6: Create real time datasets from social/ public websites using information retrieval techniques

CO7: Create efficient machine learning models to perform analysis and prediction on real datasets

MAPPING OF COURSE OUTCOMES WITH PO, PSO:

	PO'S												PSO'S	
	1	2	3	4	5	6	7	8	9	10	11	12	1	2
CO1	S			S	M								S	M
CO2	S			S	M								S	M
CO3	S		S	S									S	
CO4	S													
CO5	S													
CO6	S			S	M								S	M
CO7	S	S			M								S	M

UNIT I: Introduction to Machine Learning

Introduction, Motivation for machine learning, Applications, Abstraction and Generalization, Machine Learning Lifecycle,Types of Machine Learning, Regularization algorithms, Clustering algorithms, Deep learning al gorithms, Ensemble learning, Matching data to an appropriate algorithm, Data Retrieval, Web Scrapping, packages in Python

UNIT II: Simple Linear Regression

Regression examples, Regression models,Steps in regression analysis, Linear regression, Simple linear regressi on, Least squares estimation, Least squares regression-Line of best fit, Illustration, Direct regression method,Maximum likelihood estimation, Matrix approach,Coefficient of determination (R-squared), Example, Validation Testing Overfitting and underfitting, Logistic regression.

UNIT III: Multiple Regression and Model Building

Ordinary least squares estimation for multiple linear regression and model building, Partial regression coefficients, Standardized regression coefficients, Missing data, Validation of multiple regression model, Coefficient of multiple determination (R-Squared), Adjusted R-squared.

UNIT IV: Introduction to Classification & Classification Algo-rithms

Classification algorithms, Instance based learning, K-Nearest neighbor, Decision trees, Decision tree Bayesian algorithms, Naive Bayes Classifier, Ensemble, Bagging, Boosting, Random forests, Neural networks, Activation functions, Feed forward neural network, Multi-layer perceptron, Back prop algorithm, Recurrent or feedback architecture, Perceptron rule, Gradient-descent (training examples, Multilayer networks and back propagation algorithm, SVM and its types with implementation, ROC curves, Cost Benefit Analysis (CBA).

UNIT V: Clustering Techniques

Clustering, Clustering algorithms,Major clustering approaches, Types of clusters, Cluster centroid and distances, External criteria for clustering quality, Different aspects of cluster validation,Measures of cluster validity, Measuring cluster validity via correlation, Using similarity matrix for cluster validation, Internal measures of clustering

EXPERIMENTS:

1. Data Scraping – Frontend to Backend – BeautifulSoup

2. Multiple linear Regression – Model the strike rate of a batsman in ODI
3. k Nearest Neighbor – ODI Bowler with closest characteristics

4. Decision Tree Learning – What decides the outcome of the cricket match

5. Naive Bayes Classifier – How Bayes define the Decider

6. k Means Clustering – Cluster similar kind of players part 01

7. k Medoid Clustering – Cluster similar kind of players part 02

8. DBSCAN Clustering – Cluster similar kind of players part 03

9. Support Vector Machines – Predict Win, Loss, Draw

10. Principal Component Analysis – Act on Handwritten Characters

11. Case Study Project

TEXT BOOK(S):

1. Machine Learning, 2019 (IBM ICE)

2. Tom Mitchell, Machine Learning, McGrawHill, 1997

REFERENCES BOOKS:

1. Andrew Ng, Machine Learning, deeplearning.ai, 2018

Course Coordinator **Module Coordinator**

Programme Coordinator **HoD/CSE**